

BELLINGHAM, WA, USA

WMO: 727976

Lat: 48.794N

Lon: 122.537W

Elev: 45

StdP: 100.78

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24217

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.4	-4.0	-16.3	0.9	-3.5	-12.4	1.3	-2.0	14.6	6.5	12.9	6.4	5.3	20	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	26.8	18.5	24.9	17.7	23.1	16.8	19.1	25.4	18.2	23.9	17.3	22.3	2.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	11.8	21.5	15.9	11.3	20.6	15.1	10.8	19.5	54.5	25.6	51.4	23.9	48.7	22.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.3	8.1	-9.2	30.5	2.6	2.2	-11.0	32.0	-12.6	33.3	-14.0	34.5	-15.9	36.1		
(5)			-10.3	21.0	2.6	1.1	-12.2	21.8	-13.7	22.4	-15.1	23.0	-17.0	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.5	5.1	7.1	9.5	12.6	15.1	17.5	17.5	14.7	10.4	6.8	4.1		
	DBStd	5.58	3.92	3.22	2.77	2.41	2.52	2.24	2.19	1.99	2.29	2.63	3.66	3.81		
	HDD10.0	772	172	139	97	37	6	0	0	0	1	27	106	186		
	HDD18.3	2925	428	371	349	266	178	101	43	40	112	247	347	443		
	CDD10.0	923	2	1	6	21	86	153	232	232	141	38	9	2		
	CDD18.3	36	0	0	0	0	0	4	16	14	2	0	0	0		
	CDH23.3	350	0	0	0	1	11	52	138	122	26	0	0	0		
	CDH26.7	53	0	0	0	0	1	7	27	17	3	0	0	0		
(14) Wind	WSAvg	3.2	3.6	3.5	3.6	3.1	3.0	3.3	3.3	3.0	2.4	2.7	3.5	3.6		
(15) Precipitation	PrecAvg	910	117	86	77	70	60	46	33	37	46	83	126	120		
	PrecMax	1198	269	220	136	155	151	121	73	121	120	201	295	254		
	PrecMin	646	20	12	19	32	10	6	0	0	1	7	35	25		
	PrecStd	139	63	48	29	24	31	26	22	31	31	42	58	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	14.7	17.8	21.6	24.8	27.6	29.5	28.7	26.0	20.2	17.0	14.3		
		MCWB	11.0	9.7	11.3	13.9	16.1	18.0	19.8	19.2	17.3	13.8	12.6	11.0		
	2%	DB	12.4	12.5	14.9	18.0	22.0	24.7	26.8	26.3	23.5	17.9	14.4	12.2		
		MCWB	9.8	8.6	9.9	11.9	14.9	17.5	18.9	18.6	16.6	13.2	11.7	9.5		
	5%	DB	11.1	11.1	13.0	15.9	19.7	22.1	24.6	24.4	21.7	16.2	13.0	10.4		
		MCWB	8.9	7.9	9.3	10.9	14.1	16.0	17.9	17.7	16.0	12.5	10.7	8.2		
	10%	DB	9.9	9.9	11.7	14.0	17.6	20.0	22.7	22.6	19.7	14.8	11.9	9.1		
		MCWB	7.9	7.2	8.3	9.7	13.1	14.8	16.8	16.9	15.1	11.8	9.9	7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.2	12.0	14.9	16.9	19.3	21.0	20.2	18.7	15.8	13.6	11.8		
		MCDB	13.3	13.3	15.8	20.3	22.8	26.1	27.7	26.8	24.0	18.5	15.5	13.7		
	2%	WB	10.3	9.6	10.8	12.8	15.6	17.7	19.3	19.1	17.5	14.2	12.2	9.9		
		MCDB	11.9	11.4	13.9	16.9	20.8	23.8	25.8	25.3	22.3	16.4	13.9	11.8		
	5%	WB	9.1	8.5	9.8	11.5	14.6	16.3	18.2	18.1	16.5	13.1	11.1	8.4		
		MCDB	10.7	10.4	12.5	14.7	18.9	21.3	23.8	23.7	20.6	15.5	12.8	10.0		
	10%	WB	7.9	7.6	8.8	10.4	13.6	15.2	17.2	17.2	15.6	12.2	9.9	7.4		
		MCDB	9.5	9.5	11.3	13.3	17.2	19.3	22.1	22.0	18.9	14.2	11.7	8.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.1	7.5	8.1	8.4	8.1	9.0	9.4	9.3	7.9	6.6	5.7		
		MCDBR	7.2	9.0	10.3	12.1	13.0	13.0	13.3	13.3	13.3	10.1	7.2	6.3		
	5% WB	MCWBR	4.9	5.8	6.2	7.0	7.2	6.7	6.4	6.6	7.2	6.2	4.7	4.5		
		MCWBR	6.5	7.3	8.9	10.5	11.3	11.8	12.4	12.4	11.6	8.6	6.4	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.310	0.327	0.353	0.376	0.354	0.349	0.354	0.338	0.332	0.317	0.308		
		taud	2.517	2.554	2.519	2.433	2.381	2.456	2.496	2.493	2.531	2.539	2.523	2.493		
	Ebn at Noon		758	857	893	898	885	906	905	886	869	814	746	713		
		Edh at Noon	61	72	87	105	115	108	102	98	85	72	59	55		
(44) All-Sky Solar Radiation	RadAvg		0.90	1.77	2.73	4.18	5.39	5.85	6.45	5.45	3.84	2.11	1.07	0.72		
	RadStd		0.13	0.25	0.34	0.41	0.44	0.61	0.50	0.38	0.32	0.26	0.18	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	+13
(47) Regional (72 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page